

Sector-Wise Analysis and Comparison of S&P 500 Company Stock Prices

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Abstract—This paper examines sector-wise behavior within the S&P 500 index, aiming to detect performance differences and predict future cumulative returns. Using an open-source dataset of historical price and market capitalization data for S&P 500 constituents, daily weights are reconstructed and back-propagated market caps to form a detailed sector-level time series. An LSTM-based neural network architecture forecasts each sector’s cumulative returns under various coverage thresholds, highlighting how missing data or index membership changes impact results. The findings infer that certain sectors deviate considerably from the broader index while other maintain a constant trend.

I. INTRODUCTION

The equity markets have been a cornerstone and key indicator of economic health and financial sentiment for well over one hundred years. Among these markets, the S&P 500 has stood out as one of the most well-known and extensively researched indices. Founded in 1957 by Standard & Poor’s [1], this index comprises 500 of the largest publicly traded companies in the United States by market capitalization. As such, this index contains a large number of historical data and data points with a wide range of sector and industry coverage. As such, it has evolved into a key reference point for both retail and institutional investors as well as many academics and policymakers. Fluctuations on this index are often utilized to gauge fluctuations in the overall US economic performance [1].

The Global Industry Classification Standard (GICS) classifies listed companies into eleven sectors, ranging from defensive (Consumer Staples, Utilities) to cyclical (Information Technology, Consumer Discretionary). While the S&P 500 represents a weighted average of the top 500 U.S. companies, sector-level returns can differ significantly. This paper compares historical sector performance to the overall S&P 500 using daily or monthly closing prices for sector-based ETFs or aggregated sector indices.

the core objectives of this data analytics task are:

- Provide sector-wise performance comparisons over various market phases (bullish and bearish).
- Highlight persistent sector outperformance or underperformance relative to the S&P 500.
- Demonstrate time-series forecasting for future sector performance, volatility, and risk.

II. BACKGROUND

A. S&P 500 Index

As stated above, the Standard & Poor’s 500 (S&P 500) is a market capitalization weighted index of leading US corporations [1]. Its weighted structure ensures that larger companies exert a more pronounced influence on the index’s overall movement.

Historically, the S&P 500 has exhibited long-term growth with periodic corrections [1]. Iconic downturns—such as the dot-com bust in 2000, the Global Financial Crisis of 2008, and more recent pandemic-related volatility—have shown how external shocks can rapidly reshape market sentiment. Despite these episodes, many investors treat the S&P 500 as a go-to measure of broad economic well-being, referencing it in discussions of consumer confidence, investment strategy, and even policy-making decisions.

B. Sector Classification

While the S&P 500 as a whole offers a panoramic view of corporate America, sector-wise analysis provides greater nuance. The Global Industry Classification Standard (GICS) categorizes each listed firm into a main sector based on its principal business activities [3]. Presently, the recognized sectors include:

- Communication Services
- Consumer Discretionary
- Consumer Staples
- Energy
- Financials
- Health Care
- Industrials
- Information Technology
- Materials
- Real Estate
- Utilities

Understanding sector performance is critical for multiple reasons. First, sector cycles often lead or lag broader economic cycles. For example, the Consumer Discretionary sector tends to perform well in expansionary periods when consumer spending is high, whereas sectors like Consumer Staples and Utilities, which provide essential goods and services, are more defensive [4]. Second, sector-specific catalysts can generate enormous gains (or losses). A sharp rise in commodity prices

might boost Energy or Materials, while breakthroughs in artificial intelligence could fuel Information Technology stocks.

- **Sector-wise breakdowns can:**

- **Identify Leading and Lagging Indicators:** Certain sectors can act as under-perform for the broader economy. For instance, strong performance in Materials might signal increasing industrial demand, while a slowdown in Financials could foreshadow credit or liquidity issues.
- **Enhance Risk Management:** By evaluating which sectors exhibit higher volatility, investors can balance their exposure according to risk tolerance and market outlook.
- **Optimize Asset Allocation:** Whether for hedge funds, pension funds, or individual portfolios, understanding sector-specific performance can help optimize asset allocation.

C. Time Series Forecasting

Time series forecasting is a cornerstone of financial analysis, and its methods have shifted significantly over time, moving from traditional approaches to machine learning techniques. Classic models like ARIMA (AutoRegressive Integrated Moving Average) and VAR (Vector Autoregression) have long been used to uncover patterns and relationships within price or return data. Meanwhile, models such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity) have proven particularly useful for analyzing volatility, which is especially critical in sectors like Energy or Technology. In recent years, however, machine learning and deep learning have taken center stage in forecasting. Techniques like LSTM (Long Short-Term Memory) networks have gained traction for their ability to handle complex, non-linear data and uncover intricate patterns that older models might miss. That said, their effectiveness often depends on factors like access to powerful hardware and the specific needs of stakeholders, making their application more nuanced and context-dependent.

III. HYPOTHESIS

A. Hypotheses

The two main components of this hypotheses are:

- 1) **Sector Divergence:** Being able to identify trends and visualize that not all sectors perform proportionally to the S&P index, with many diverging
- 2) **Predictive Value of Time Series Models:** Accurately train and forecast sector-wise performance

IV. THE DATA

A. Dataset Properties

The data set gathered is an open source data set [2] from kaggle.com that contains three separate datasets, one of the S&P 500 index price over time, another of each stock in the index over time, and lastly a dataset containing company information such as market cap, trading volume, company sector and industry, and price for that particular timestamp.

This dataset therefore represents specific attributes in relation to the S&P 500 index, making data analysis a very important and applicable task.

As stated, the 3 datasets contain a variety of potential features. The first csv contains:

Index	Feature	Datatype
0	Date	Object
1	S&P500	float64

TABLE I
DATASET FEATURES

The most crucial csv, the individual stock performance overtime contains features:

Index	Feature	Datatype
0	Date	Object
1	Symbol, Col 2	Object
2	Adj Close	float64
3	Close	float64
4	High	float64
5	Low	float64
6	Open	float64
7	Volume	float64

TABLE II
SP500_STOCKS.CSV CONTENTS

The Symbol feature represents the stock symbol (or ticker) [1] and is the same as in the sp500_companies.csv (see below) the 'Adj close' price feature represents the adjusted closing price of a stock. This represents the same as the close price but also also takes into consideration company actions such as dividends and share-splits [1]. As such this will be the close price used for this experiment.

The sp500_companies.csv contains far more columns as shown below:

Index	Feature	Datatype
0	Exchange	Object
1	Symbol	Object
2	Shortname	Object
3	Longname	Object
4	Sector	Object
5	Industry	Object
6	Currentprice	float64
7	Marketcap	int64
8	Ebitda	float64
9	Revenuegrowth	float64
10	City	Object
11	State	Object
12	Country	Object
13	Fulltimeemployees	float64
14	Longbusinesssummary	Object
15	Weight	float64

TABLE III
SP500_STOCKS.CSV CONTENTS

B. Data Pre-processing

One of the key stages for any data analysis task is data pre-processing. This involves cleaning and organizing the data into such a manner that it is most optimized to be processed via a predictive model or in such a manner that it can be effectively visualized. Data pre-processing can involve many

stages, such as datatype conversions, null value handling, and dimensionality reduction, all of which are crucial to allow a model to make accurate predictions. Furthermore, by analyzing the dataset, it is possible to remove redundant data and reduce the required complexity.

Data pre-processing steps for this experiment were mainly focused on the `df_stocks.csv` as this is what the time series analysis is performed on. The other datasets are mainly used for a cross comparison and visualizations of potential trends.

1) *Type conversions and handling*: Ensuring that all features are in the correct format is a critical first step for effective time series modeling. Financial datasets often contain columns that are either mislabeled or stored as generic string (object) types when numerical types (float or integer) would be more appropriate. Moreover, date columns are frequently provided as strings rather than specialized date-time objects, which impedes the ability to index and segment the data in time-based analysis. Considering null or missing values in stock market data is also crucial, these can arise for various reasons, including market holidays, ticker symbol changes, or corporate actions not recorded in the dataset. These missing data points can lead to errors or biases in subsequent analyses or modeling if not handled appropriately.

Firstly the 'date' column in the `df_stocks.csv` was converted into date-time format in order to be compatible for model prediction. Following this, the number of null data-points for the given symbols within the dataset were identified as 100475. Upon further investigation it is revealed that this is because as a result of the changing companies in the S&P index, many companies have not been in the index for long, with it being that the first date where all companies have a non-null value being 2024-09-24, with the earliest date over half the companies had values in the dataset being 2010-01-04. This also shows a potential limitation in the dataset, as it does not take into account companies that were dropped from the S&P 500 overtime. However, as this experiment completes a sector-wise analysis of the S&P 500 performance as opposed to purely individual stocks, the lack of early data should not hinder performance too much as a sector-wise average is performed for results analysis.

2) *Calculating Share-count and Back-propagation of Market-cap and Weights*: A "reference date" (e.g., 2024-12-20) is adopted to fix each stock's final-day market cap and adjusted price, and which derives a constant share count:

$$\text{Shares}_i = \frac{\text{MarketCap}_i(\text{ref})}{\text{Price}_i(\text{ref})}.$$

Despite real historical changes, this snapshot-based approach keeps each stock's shares consistent across all dates. We then compute a back-propagated market cap on any day t :

$$\text{BackpropMarketCap}_i(t) = \text{Shares}_i \times \text{AdjClose}_i(t).$$

Grouping by date and summing yields the day's total back-propagated cap, letting us track how much of the index or

sector each stock occupies. Daily weights:

$$\text{Weight}_i(t) = \frac{\text{BackpropMarketCap}_i(t)}{\text{TotalBackprop}(t)},$$

capture a stock's percentage share of the whole. This helps form sector-level returns, as larger stocks (by cap) influence overall performance more. Such back-propagation is crucial because (1) historical share counts are rarely available, and (2) we need a consistent per-day measure to reflect each stock's contribution in a comparable index or sector framework.

Though it may slightly diverge from reality if a company significantly changed its shares outstanding in the past, this approach remains a common practical workaround. It enables consistent weighting logic across days and symbols, forming the backbone for any weighted return calculation or forecasting at a sector or index level.

3) *Experiment Details*: Traditional models like ARIMA and VAR capture only part of stock-price behavior, often failing to handle the nonlinear dynamics introduced by external shocks or sector-specific events. In contrast, Long Short-Term Memory (LSTM) networks employ gating mechanisms to manage extended sequences [5]. LSTMs also handle the cyclical, seasonal, and trend aspects of financial data, potentially outperforming simpler approaches by incorporating features like volume or volatility. LSTM was chosen for its capacity to capture both long-range and short-term dependencies.

Our time-series experiment uses weighted-average Percentage Cumulative Daily Returns, examining multiple time periods and coverage thresholds (e.g., 90%, 95%, 100%). These thresholds address the fact that some stocks joined the index later or have incomplete data. To ensure valid samples, we identified the earliest date from which each coverage level had sufficient non-null values for subsequent analysis, as illustrated below.

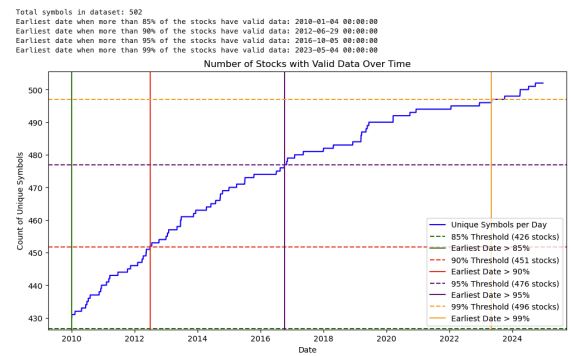


Fig. 1. Time-windows

As such, coverage thresholds of 90%, 95%, and 100% are adopted to handle missing data, selecting dates and symbols with sufficient completeness. We then compute *weighted average cumulative returns* by sector under these filtered conditions.

a) *Daily Percentage Changes*: First, we calculate each stock's daily percentage change from its adjusted close price:

$$r_i(t) = \frac{P_i(t) - P_i(t-1)}{P_i(t-1)},$$

where $P_i(t)$ is the stock's adjusted close on day t . This normalization captures each stock's relative movement while accounting for dividends and splits, enabling comparisons across firms of varying share prices or market caps.

b) *Weights and Re-normalization*: Next, we assign a *weight* to each stock based on market capitalization (e.g., final-day share count multiplied by the daily price) or a similar metric. If $w_i(t)$ is that raw weight, we re-normalize them to sum to 1 within each sector on each date:

$$w_i^{\text{sector}}(t) = \frac{w_i(t)}{\sum_{j \in S} w_j(t)},$$

where S represents all stocks in the sector. This step ensures the aggregated sector return reflects sector specific proportions rather than an index wide reference.

c) *Sector's Daily Return*: Given these re-normalized weights and daily returns, the sector's daily return is:

$$R_{\text{sector}}(t) = \sum_{i \in S} \left(w_i^{\text{sector}}(t) \times r_i(t) \right).$$

Grouping the data by *sector* and *date* ensures each day's sector-level return involves only valid, active stocks.

d) *Cumulative Return*: Finally, we compound these daily returns into a cumulative measure:

$$\text{Cumulative}(T) = \prod_{t=1}^T (1 + R_{\text{sector}}(t)) - 1.$$

This value indicates how much a sector has grown or declined from the chosen start date.

Architecture and Layer Choices:

- **LSTM Layers (Two Stacked)**: Two LSTM (Long Short-Term Memory) layers are used for the model. The first LSTM layer has `return_sequences=True`, (it outputs a sequence of hidden states for each timestep, feeding these to the next LSTM layer). The second LSTM layer has `return_sequences=False`, extracting the final hidden state. The first layer captures short-term dependencies, while the second layer can model more refined patterns on top of those features. Each LSTM layer has 50 units, which is a balanced choice for capturing complexity without excessively large parameter sets.
- **Dense Layers**: After the second LSTM, we have two dense (Dense) layers: one with 25 units and the final output layer with a single unit. The penultimate Dense layer provides additional learnable parameters to transform the LSTM's final hidden representation, while the last Dense(1) layer produces a single scalar for the daily cumulative return forecast. Essentially, this structure turns the final LSTM encoding into the next predicted value.
- **Optimizer and Loss**: The model is compiled with the 'adam' optimizer and a 'mean_squared_error'

loss. The Adam optimizer is widely used in time-series tasks due to its robust and efficient convergence properties, while MSE is a standard regression loss function that penalizes large prediction errors more severely.

Scaling and Data Preparation:

- **MinMax Scaling**: Each sector's `Cumulative_return` series is scaled into the range $[0, 1]$ via the `MinMaxScaler`. This step aids LSTM training by keeping numeric values bounded and reducing issues with very large or small magnitudes. Once the model produces its predictions in the scaled domain, we invert that scaling to get the forecasted return values back in the original range.
- **Sequence Generation**: We define a `window_size` of 60, so that 60 consecutive past timesteps are used to predict the next day's cumulative return. Mathematically, if r_t is the scaled return on day t , we form:

$$X_i = (r_{i-60}, r_{i-59}, \dots, r_{i-1}),$$

$$y_i = r_i.$$

This approach helps the LSTM learn temporal patterns over approximately two trading months. Using a shorter window may lose the ability to capture slower-moving cycles and longer window could risk overfitting.

Rolling Forecast Logic:

- **Iterative Prediction**: After training, the script performs an iterative, "rolling" forecast for `future_days` steps (by default, 356). We take the last 60 points of scaled data, pass them into the LSTM to obtain the next forecasted point, then *append* that forecast as if it were real data to slide forward by one timestep. This repetition yields a multi-step forecast horizon.
- **Inverse Scaling**: Once all scaled predictions are produced, we invert them back to the original scale using `scaler.inverse_transform(...)`. This step is key for interpretability: it ensures the final values reflect actual proportions or levels of `Cumulative_return` rather than dimensionless scaled numbers.

Choice of 60-day Window & 356-day Forecast: Using `window_size=60` is motivated by the idea that roughly 2-3 months of past data is a reasonable horizon to capture short and mid-term trends in stock returns. Financial time series often exhibit patterns or memory in the range of weeks to months. A smaller window could miss important slower-moving influences, while a larger window could add noise or require excessive computational resources. Forecasting 356 days (nearly one trading year) offers a balanced look at longer-horizon trends.

4) *Results and Analysis*: Below are the experimental results:

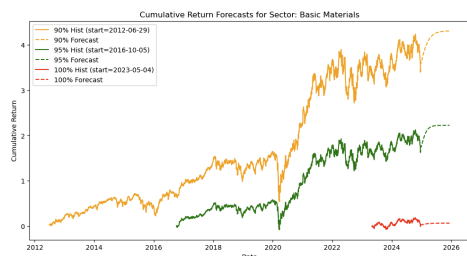


Fig. 2. Basic Materials

From the table, the yearly total forecasted increase is 0.017675 under 100% coverage, 0.668992 under 90%, and 0.449616 under 95%. The historical lines and short-term forecasts suggest that the 90% scenario projects the highest net gain, while the 100% coverage scenario stays relatively flat. Overall, Basic Materials sees moderate-to-high growth potential in the 90% coverage subset, but more subdued gains under 100%.

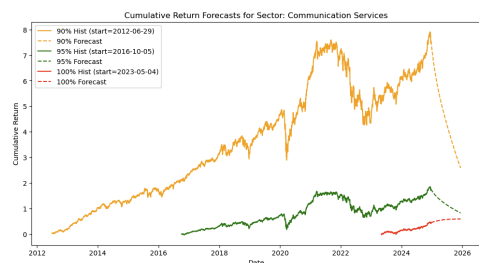


Fig. 3. Communication Services

Here, the table shows +0.131434 for 100%, -4.967217 for 90%, and -0.939691 for 95%. The 90% coverage yields a strongly negative forecast, while 100% coverage remains slightly positive. This disparity could reflect different symbol sets included in each coverage window.

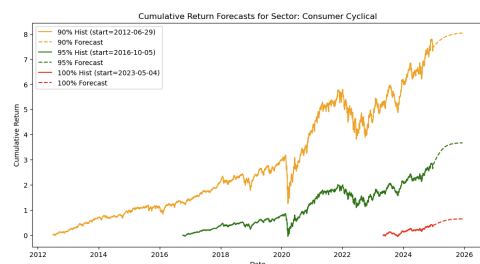


Fig. 4. Consumer Cyclical

According to the table, the 100% coverage scenario forecasts +0.247724, 90% yields +0.480055, and 95% climbs to +0.844933. Notably, the 95% coverage subset projects the highest overall increase, suggesting that excluding some missing-data stocks might focus on stronger performers.

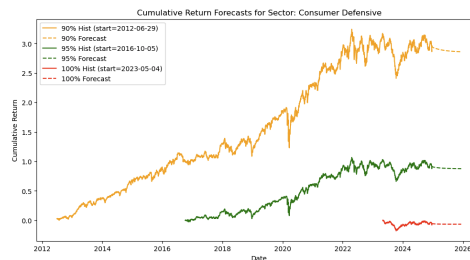


Fig. 5. Consumer Defensive

Table values are -0.016592 for 100%, -0.106707 for 90%, and -0.047638 for 95%. All coverage levels yield slightly negative increases in the forecast, implying that, under this model, Consumer Defensive might pull back over the chosen horizon. The historical lines show strong growth into 2022–2023, but the forecast lines dip gently afterward, consistent with a mild downward outlook across all coverage thresholds.

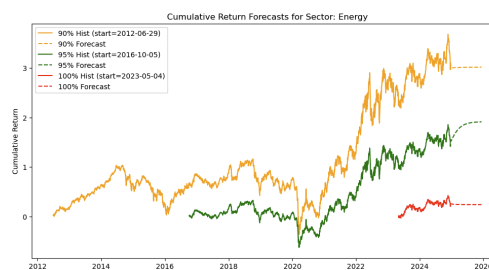


Fig. 6. Energy

Yearly total increase from the table is -0.031328 for 100%, -0.093539 for 90%, and +0.353197 for 95%. The 95% subset stands out with a positive forecast, whereas the 90% and 100% subsets predict mild declines. Historically, Energy shows large swings, and the model's forecasts reflect these potential divergences: the green line (95%) suggests an upside that might be tied to excluding certain lagging stocks.

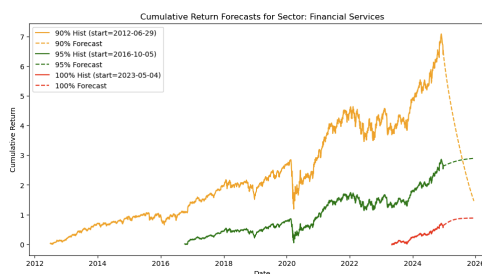


Fig. 7. Financial Services

From the table, we see +0.231982 for 100%, -5.031353 for 90%, and +0.219270 for 95%. The 90% coverage scenario forecasts a strongly negative outcome, while both 95% and 100% scenarios stay modestly positive. This discrepancy likely arises from key financial symbols that are missing data in the narrower 90% window.

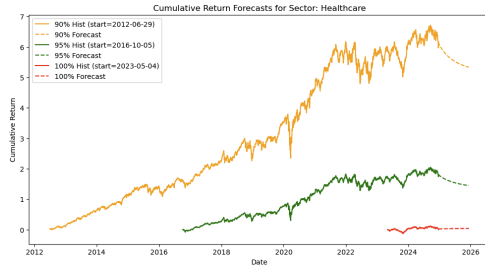


Fig. 8. **Healthcare**

The table lists +0.008846 for 100%, -0.834560 for 90%, and -0.375337 for 95%. Overall, Healthcare's final forecast is near neutral or negative in the 90%/95% scenarios, but slightly above zero for the 100% coverage. Historically, healthcare has trended upwards significantly, yet the model's short-term forecast indicates a mild downturn in some coverage sets. The difference in coverage sets underscores that the underlying composition of stocks can shift the performance outlook.

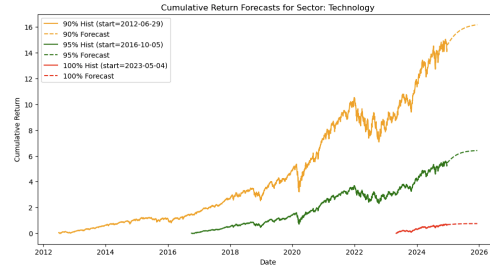


Fig. 11. **Technology**

The table results are +0.071060 for 100%, +1.489273 for 90%, and +0.921669 for 95%. Technology stands out with large positive forecasts across all coverage scenarios, reflecting both its robust historical growth and strong short-term predictions. The 90% scenario in particular might exclude certain slower-moving or incomplete-data symbols, leading to an even higher aggregated forecast.

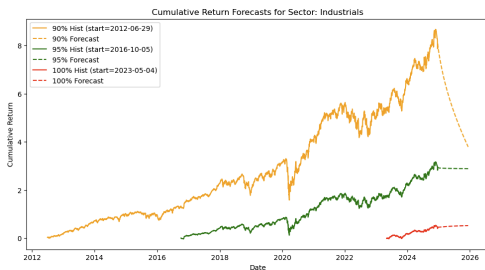


Fig. 9. **Industrials**

From the table, 100% yields +0.050362, while 90% is -4.283737 , and 95% is -0.083041 . Similar to some other sectors, the 90% subset includes stocks that the model anticipates a significant drop for. Meanwhile, 100% coverage barely remains positive, and 95% is slightly negative.

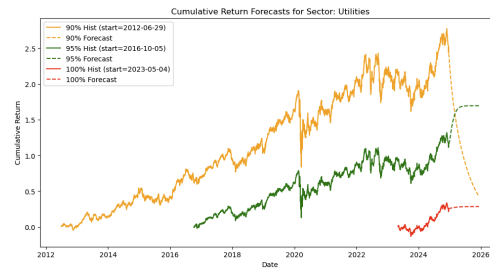


Fig. 12. **Utilities**

From the table, 100% coverage sees +0.018545, 90% is -2.003159 , and 95% is +0.524365. Utilities' 90% coverage subset shows a noticeable decline, whereas the 95% and 100% sets maintain positive forecasts.

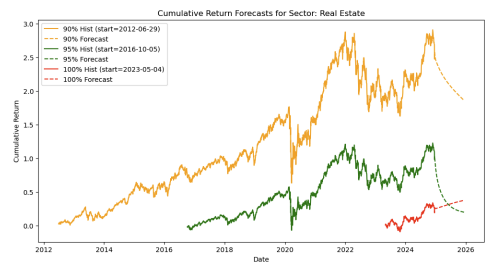


Fig. 10. **Real Estate**

The table shows +0.103329 for 100%, -0.717837 for 90%, and -0.806357 for 95%. While 100% coverage suggests mild positive growth, the 90% and 95% sets slip into fairly negative territory. The large sector-level divergence might indicate that missing data or specific real-estate stocks with incomplete histories carry heavily negative forecasts.

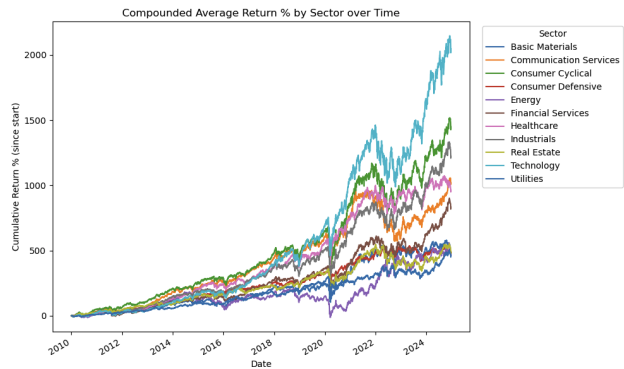


Fig. 13. **Compounded Returns Over Time**

TABLE IV
YEARLY TOTAL FORECASTED INCREASE BY SECTOR & COVERAGE

Sector	100%	90%	95%
Basic Materials	0.017675	0.668992	0.449616
Communication Services	0.131434	-4.967217	-0.939691
Consumer Cyclical	0.247724	0.480055	0.844933
Consumer Defensive	-0.016592	-0.106707	-0.047638
Energy	-0.031328	-0.093539	0.353197
Financial Services	0.231982	-5.031353	0.219270
Healthcare	0.008846	-0.834560	-0.375337
Industrials	0.050362	-4.283737	-0.083041
Real Estate	0.103329	-0.717837	-0.806357
Technology	0.071060	1.489273	0.921669
Utilities	0.018545	-2.003159	0.524365

Overall, performance differed radically sector to sector, with some forecasts predicting steep increases while others steep decreases. From this limited experiment it could be hypothesized that for quite a few of the measured forecasts, the LSTM did not generalize well in training as the forecast appears inconsistent with historic price data in a few instances

V. FUTURE WORKS

In this study, only stock prices over time were available, so the analysis focused on computing sector-level cumulative returns relative to the starting date while back-propagating individual stock weights and marketcap. While this approach provides a reasonable performance metric it would be ideal to have actual historic data instead of requiring back-propagating. In future research, having access to a dataset containing the number of shares over time or the historical market caps of each constituent would allow for a more direct analysis of true sector valuations.

The current implementation employed a multi-layer LSTM network for forecasting. However, testing various combinations of architectures (e.g., deeper LSTM stacks, GRU layers, Transformer-based models) and hyperparameters could lead to more accurate and robust predictions. In addition, comparing neural networks against classical statistical methods (like ARIMA, VAR) and traditional machine learning approaches (e.g., Random Forest, Gradient Boosting) would reveal the relative strengths and weaknesses of each method. A thorough cross-comparison ensures that the most appropriate model—both in terms of predictive power and computational feasibility—can be deployed confidently in real-world applications.

Aswell as choosing an overarching model, tuning hyperparameters (learning rates, number of units in each LSTM layer, dropout rates) is critical for maximizing performance. Techniques such as grid search, random search, or Bayesian optimization could be employed to systematically explore parameter spaces. This step would be particularly valuable if deeper or alternative network architectures are used; understanding how changes in layer depth or width influence overfitting, convergence speed, and final accuracy can yield practical guidelines for constructing high-performing time-series models.

VI. CONCLUSION

This study demonstrates a sector-wise analysis of S&P 500 companies by back-propagating each stock's market cap from a final reference date and aggregating daily weights to produce sector-level returns. By training an LSTM model on these time series, we explore how different coverage thresholds—tied to data availability—shape both historical and forecasted performance. The results confirm that sectors diverge meaningfully: some are more volatile (e.g., Energy, Financial Services), while others exhibit more defensive behavior. The comparative forecasting exercise also underscores the influence of re-normalized weights in capturing authentic sector trends. Going forward, the integration of additional features (e.g., macroeconomic indicators, fundamental data) could further refine model accuracy.

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